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THE SAME JOURNEY, ONE LAYER UP EACH TIME



Death by PowerPoint

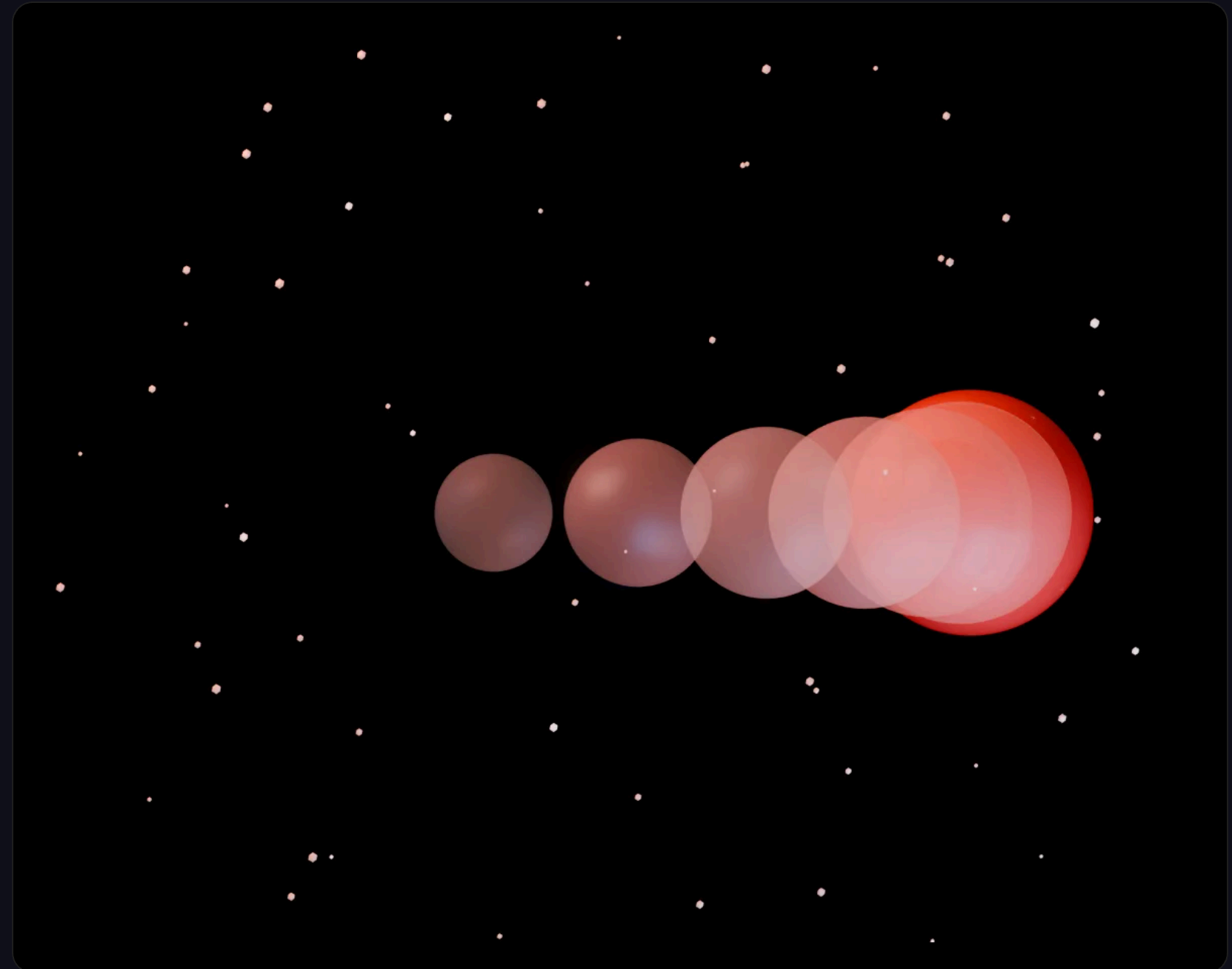
coined by Angela R. Garber · **Small Business Computing, 2001**

twenty-five years of advice aimed at the presenter

</> LOOK CLOSER

This is not a slide.
It's a **web page.**

 sunholo.com/presentations/death-by-powerpoint/



⇌ THE SHIFT

The constraint that made PowerPoint sensible is **gone**



1987 - 2024

A human places every element by hand

So give them a canvas and some boxes. Given that constraint, PowerPoint is a *good design*. Everything people complain about follows from it.



NOW

The author can write code

Once the thing assembling your deck writes code fluently, insisting it produce a proprietary binary is a strange thing to ask for.

© THESIS

Generate the **medium,**
not the ~~file format.~~

↻ THE ROUND TRIP

Count the conversions



Idea

—



.pptx

boxed into a rectangle



Their projector

fonts substitute



Export to PDF

animation flattens



Screenshot

links die



Someone's phone

unreadable

👉 ASK FOR WHAT IT'S GOOD AT

Same slide title, **two formats**

📄 .PPTX — A ZIP OF XML

```
<p:sp>
  <p:nvSpPr>
    <p:cNvPr id="2" name="Title 1"/>
  </p:nvSpPr>
  <p:spPr><a:xfrm>
    <a:off x="838200" y="365125"/>
    <a:ext cx="10515600" cy="1325563"/>
  </a:xfrm></p:spPr>
  <p:txBody><a:p><a:r>
    <a:t>Generate the medium</a:t>
  </a:r></a:p></p:txBody>
</p:sp>
```

Coordinates in English Metric Units. Written blind — the model can't open it and look.

</> HTML — THE MEDIUM ITSELF

```
<h1 class="slide-title">
  Generate the medium
</h1>
```

Renders instantly. You can see it, so the agent can be told it's wrong.

© THE FINAL MEDIUM

One arrow



IDEA



NO CONVERSION



THE SCREEN

What you gain, what you give up

+ GAINED

- ✓ No licence, no “created in a newer version”
- ✓ No font substitution — the fonts ship with the page
- ✓ Opens on anything with a browser
- ✓ Deployable to a URL, diffable in git
- ✓ The slide can *run things*

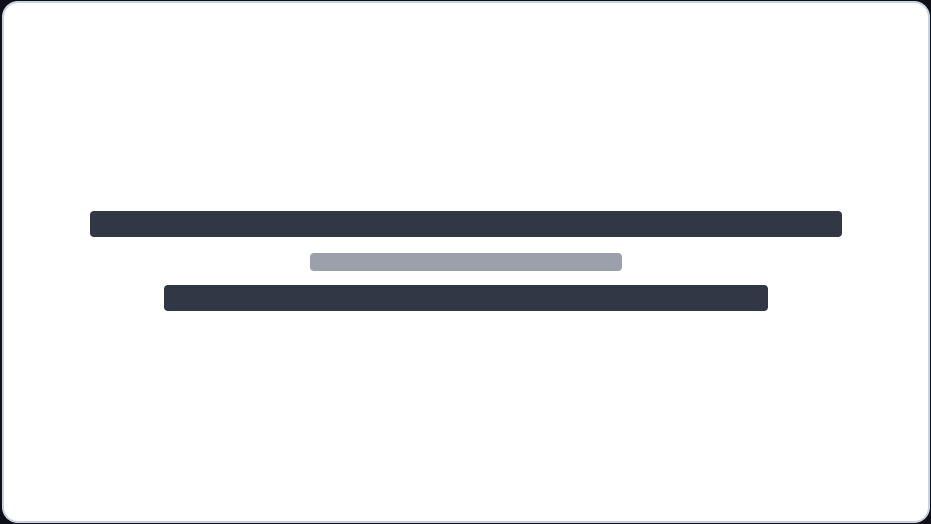
— GIVEN UP

- ⚠️ Corporate template compliance — no answer. Sometimes you lose this one.
- ⚠️ A file a colleague can edit in PowerPoint — they edit the page instead
- ⚠️ Animation by menu — you describe it instead
- ⚠️ Needs a browser, and a network for the live parts — PDF export covers the room with no wifi

🔗 THE REAL PROBLEM

Ask three times, get **three decks**

× SAME PROMPT, NO DESIGN SYSTEM



✓ SAME PROMPT, DESIGN SYSTEM IN THE REPO



 GUIDANCE, IN ENGLISH

The design system is a **document**, not a config file

 EXCERPT · .CLAUDE/SKILLS/PRESENTATION-SLIDES/SKILL.MD

-  **Orange accent** (#e73c17 dark / #d4300f light) for emphasis, **never overused**.
- **Montserrat** for all text, **JetBrains Mono** for code and data.

“Never overused” is not a number. It’s a **judgement** — and judgement is exactly what the model is good at. So write it in the language judgement comes in.

📏 ONE RULE, MOST OF THE VALUE

Never measure against the window. Measure against the **slide**.

✗ **WRONG** – sized against the browser window

```
max-height: calc(100vh - 320px); /* guesses the chrome height */
```

On a tall window the slide's content runs off the bottom of the stage. No scrollbar appears. You find out on the projector.

✓ **RIGHT** – sized against the 16:9 stage

```
max-height: 70cqb; /* 70% of the slide's height */
```

Every deck is a fixed 16:9 stage, letterboxed to fit whatever it's shown on. Design once; it looks the same everywhere.

🛡️ KEEPING IT HONEST

Written rules get followed most of the time

⚙️ .CLAUDE/SETTINGS.JSON

```
"PostToolUse": [{  
  "matcher": "Write|Edit|MultiEdit",  
  "hooks": [{  
    "type": "command",  
    "command": "node scripts/lint-slides.mjs"  
  }]  
}]
```

> WHAT THE AGENT SEES

```
$ node lint-slides.mjs
```

✗ Slide stage-safe sizing violations:

02-teaching-taste.html:214

rule: **calc-with-viewport**

line: max-height:calc(100vh - 320px)

fix: calc() with viewport units
double-counts chrome the stage
already excludes. Use cqb.

1 violation(s). exit 2

... agent reads this, fixes it, retries

✓ clean

 TAKE THIS BIT WITH YOU

Where to draw the **line**



Prose for taste

Anything needing judgement. “Never overused.” “One idea per slide.” Write it the way you’d brief a designer.



Linters for physics

Anything mechanical and checkable. Forbidden units. Missing alt text. Contrast ratios. Fail loudly.

👁 BEFORE I SWITCH WINDOWS

Four things to **watch for**

- 01 The prompt is **plain English**. I don't type any CSS.
- 02 The agent **reads the design system first**, before it writes anything.
- 03 The linter fires on the edit and the agent **fixes its own mistake**.
- 04 I reload the browser. **That's the deploy.**
No export. No conversion. No "save as".



LIVE • BUILDING A SLIDE

Switching to the repo

📄 THE PROMPT, VERBATIM – COPY IT OUT OF THE RECORDING

Add a slide to the death-by-powerpoint deck:
a two-column comparison of what we gain and
what we give up. Match the deck's design system.

Watching: **SKILL.md** → the agent writes HTML → the lint hook fires → reload.

🔄 THE LOOP

What just happened



ASK

One sentence of English



READ THE RULES

SKILL.md, design system, sizing rules



WRITE

HTML into a real file in a real repo



LINT

Hook runs automatically on the edit



RELOAD

The deploy

← lint fails → the agent reads the error → fixes it → retries

✂ BEYOND SLIDE SOFTWARE

Three things a slide couldn't do

EXHIBIT A • NEXT



It can be interactive

A live D3 decision-space explorer. Not a screenshot of one — I can drive it from here, and answer a question by changing the input.

```
../ai-slaves-human-masters-ida/01-entropy-explorer.htm  
l
```

EXHIBIT B • THEN



It can animate an argument

The development loop, moving. And a joke that isn't one: it's the same loop that built the slide you watched ten minutes ago.

```
../ai-slaves-human-masters-ida/03-development-loop.htm  
l
```

EXHIBIT C • THEN



It can render real data

Benchmarks the deck draws itself, rather than a picture of a chart someone pasted in nine months ago. Change the data, reload, it's correct.

```
../ai-slaves-human-masters-ida/05-benchmarks-motoko.ht  
ml
```

A talk is a **JavaScript array**

```
const PLAYLIST = [  
  { src: '../about-mark/about-mark.html',           minutes: 1 },  
  { src: '00-death-by-powerpoint.html',             minutes: 4 },  
  { src: '01-generate-the-medium.html',             minutes: 7 },  
  { src: '02-teaching-taste.html',                  minutes: 8 },  
  { src: '03-live-build.html',                      minutes: 11 },  
  { src: '04-beyond-slide-software.html',           minutes: 4 },  
  { src: '../ai-slaves-human-masters-ida/01-entropy-explorer.html', minutes: 2 },  
  { src: '../ai-slaves-human-masters-ida/03-development-loop.html', minutes: 1 },  
  { src: '../ai-slaves-human-masters-ida/05-benchmarks-motoko.html', minutes: 2 },  
  { src: '05-three-audiences.html',                 minutes: 6 },  
  { src: '06-whats-next.html',                     minutes: 3 },  
  { src: '07-steal-this.html',                     minutes: 3 },  
  { src: '../contact-mark/contact-mark.html',      minutes: 0 },  
];
```

8 written for today

5 borrowed from other talks, unmodified

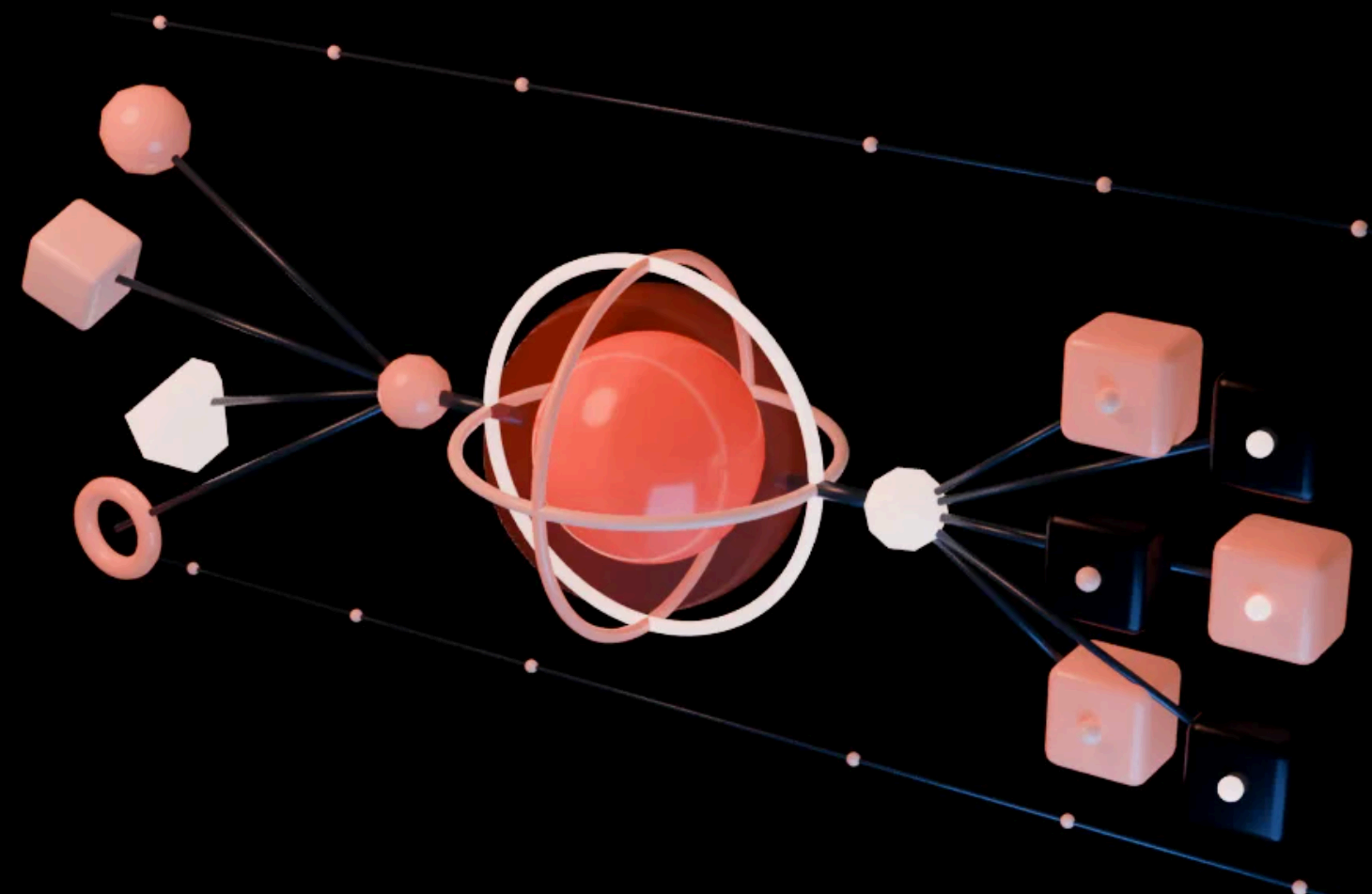
 BLENDER + HTML

A live Blender scene in the browser

The slide contains the animated model itself.

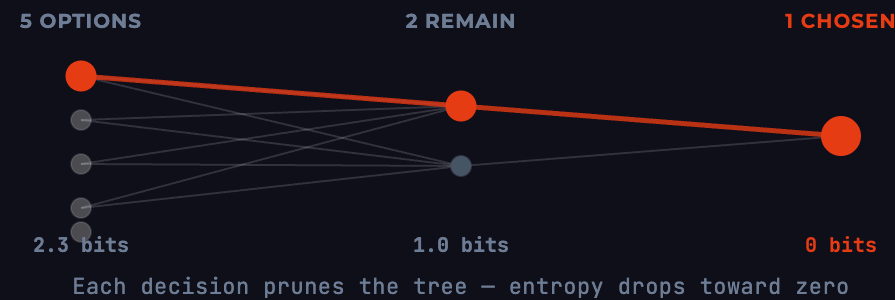
Rotate it from the stage, change the camera, or let it run as a restrained loop.

LIVE animated GLB with camera controls
PDF the same component supplies a fixed poster
LOAD the model arrives only when this slide appears



The Decision Space

Every software project starts as a vast space of **possible paths**. Each decision — a framework choice, an error strategy, an API design — eliminates alternatives and narrows what remains. **Entropy** measures how many meaningful choices are still open.



The key question isn't *how much* entropy a project has — it's *when* that entropy gets resolved. Decisions made at design time cost tokens. Decisions deferred to runtime cost debugging, deployments, and outages.

Press [→](#) to explore interactively

The Development Loop

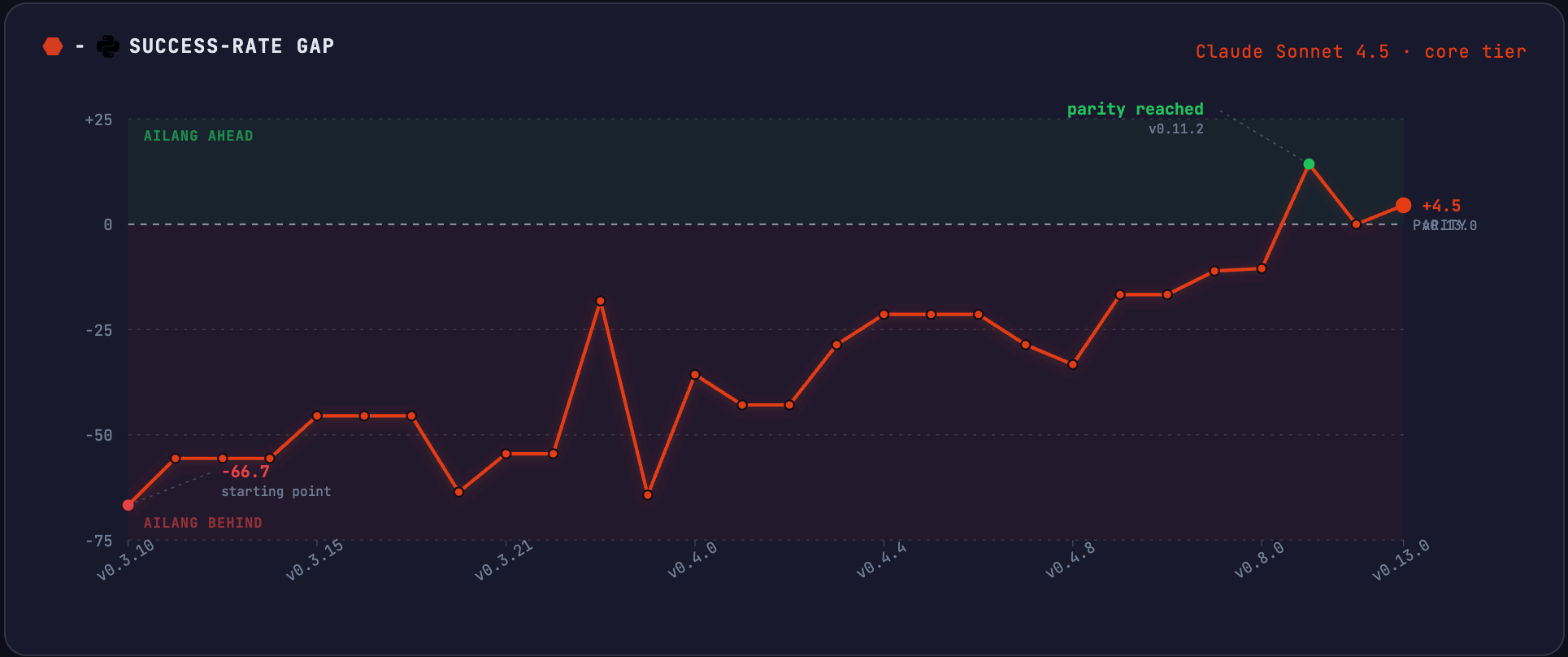
SIX STAGES · THE BRIEF LOOPS BACK · NOTHING RUNS WITHOUT IT



A human writes the brief. Nothing else runs until it's approved.

Does it **work**?

AILANG started **65 points behind** Python on Claude Sonnet 4.5. Watch the gap close.



data: ailang.sunholo.com/benchmarks/latest.json • snapshot 2026-05-11

From **-65 to parity** on Claude Sonnet 4.5. The dips near the top aren't regressions — that's where saturated benchmarks get retired and harder ones come in. The bar keeps rising even after we catch up.

📍 AILANG IN PRODUCTION

100% AI-coded apps. Running today.

Real WASM modules in your browser — no Python, no server — every extraction validated by a contract the compiler refuses to ship without.



DOCUMENT INTELLIGENCE

AILANG Parse

DOCX, PPTX, XLSX, PDF & images parsed **directly in your browser**. Track changes, merged cells, comments — preserved, not flattened.

69 files 15 formats
3 / 3 track changes (vs 0 / 5 others)

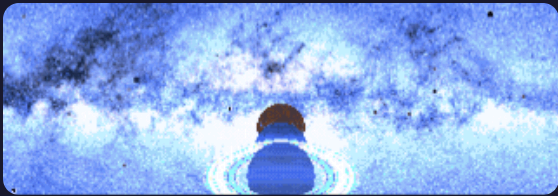


STREAMING & VOICE

Voice DocParse + Website Builder

Talk to a document via Gemini Live. Or describe a site and watch it build — AILANG WASM modules driving the whole pipeline.

WASM runtime Gemini Live bidi audio
no server



GAME ENGINE

Stapledon's Voyage

A hard-sci-fi relativity simulator. AILANG drives the **game logic** — special & general relativity, flybys, time dilation. Not a documents demo.

AILANG game core SR + GR physics
open source

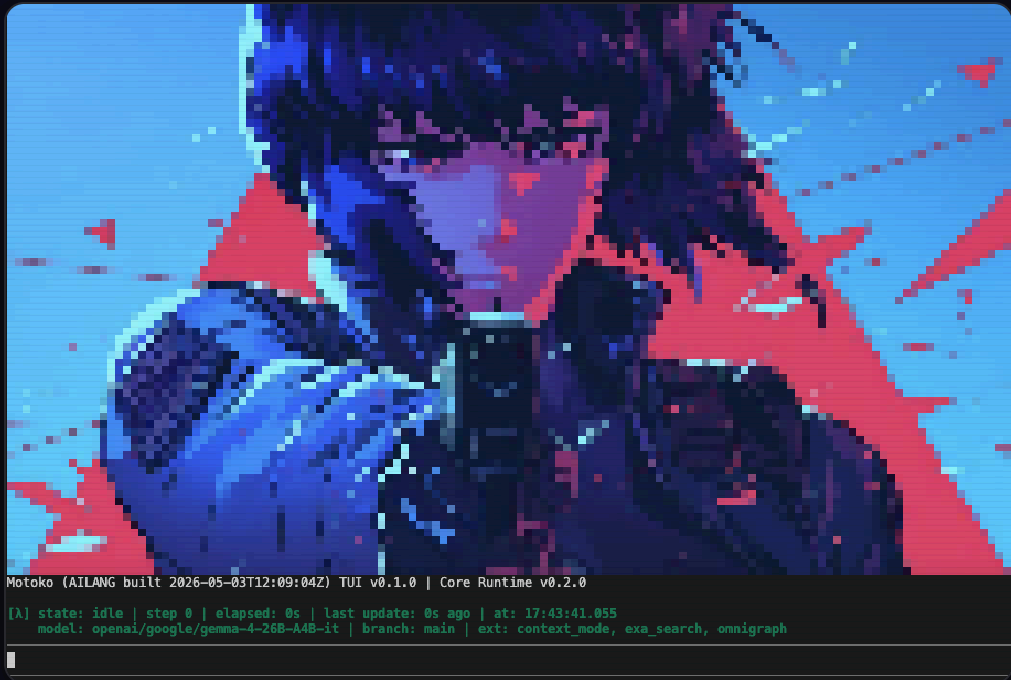
Everything you see is

100% AI-coded in AILANG

— compiler, type checker, effects, Z3 verification, WASM runtime, and stdlib included.

sunholo.com/ailang-demos

A coding harness that **grows itself** — from a verifiable core.



built by [@arniwesth](#)

λ_ motoko

Phoenix architecture

Start with a small, **verifiable** AILANG core. The agent uses the core to write more AILANG — tools, extensions, even *parts of itself*. Every step goes through the compiler. If it can't be verified, it doesn't ship.



Verifiable core

AILANG types · contracts · effects all checkable



Self-verifying loop

the agent runs its own compiler · fixes its own errors



Self-extending

AI builds new tools in AILANG · verified before they're trusted

SELF-CORRECTING LOOP

When the AI gets it **wrong**... the language tells it.

Live session · **openrouter/z-ai/glm-5** · task: wire an MCP subprocess into AILANG ·
2026-05-07

~/motoko_agent · session_2026-05-07.jsonl

step 61

AGENT Perfect! I have successfully implemented the task. ✅ Implement

no human in the loop · 8 steps of self-repair after the parse error



Don't ask AI to learn your language. Build a language AI can speak.

Cheaper to debug. Replayable. Verifiable.



STATIC TYPES • THE AI SAFETY NET

Pre-runtime, parseable errors mean one-turn fixes — not production crashes.



NEUROSymbOLIC VERIFICATION

AI writes the code, Z3 proves the contracts. Counterexamples beat tests.



WORKS IN ANY STACK

Tight loops, explicit context, structured errors — push for these where you are.



ailang.sunholo.com language • playground • demos



mcp.ailang.sunholo.com live MCP for Claude / Cursor / Cline



github.com/sunholo-data/ailang source • compiler • stdlib



github.com/arniwesth/motoko_agent the motoko coding harness

👥 ONE DECK, THREE READERS

You are writing for **three people**



The room, right now

Wants a sparse slide behind a talking human. Six words and a picture. Anything more competes with you.

USUALLY

This one you get right — it's the only reader most decks are designed for.



Someone opening the link in six weeks

Wants prose that explains itself. There is no narrator. They weren't there, and they never will be.

USUALLY

They get a deck of orphaned bullet points that meant something once.



“Can you send me the slides?”

Wants one file they can keep, forward, and search. Probably wants to print two of the pages.

USUALLY

A 40MB attachment that bounces, or a link to a login screen.

DENSITY

The same slide, **twice**

STAGE

-66.7 → +4.5

That's AILANG's LoCoBench delta against Python between v0.6 and v0.9 — the point where an AI-first language stopped being a toy.

What the projector shows. You supply the sentence out loud.

GUIDE

-66.7 → +4.5

That's AILANG's LoCoBench delta against Python between v0.6 and v0.9 — the point where an AI-first language stopped being a toy.

What the link shows a stranger. The sentence is written down.

```
/* the entire hiding mechanism */
html[data-density="stage"] .detail { display: none }

// and the line that picks the default for you
root.setAttribute('data-density', embedded ? 'stage' : 'guide');
```

press **D** now — every slide in this talk has both

The PDF is **generated**, not exported

- ✗ **It can't be a print stylesheet.** Slides are stacked absolutely with only one visible, revealed by animation from `opacity:0`, and some build up over several presses. A static print captures none of that.
- ✓ **So drive the real deck.** Headless Chrome opens it, walks every slide to its final revealed state, screenshots at 1920×1080, and assembles the pages.
- ✓ **Order comes from the same PLAYLIST as the live talk** — so the PDF cannot drift from what you actually presented. That's not a nice-to-have; it's the thing that makes it survive.

```
$ node export-pdf.mjs death-by-powerpoint --density=guide  
→ death-by-powerpoint.pdf — with the take-home prose included
```

 WHERE IT LIVES

The deck doesn't get attached.
It gets **deployed.**



sunholo.com/presentations/death-by-powerpoint/

🎤 WHAT WE MAY DO NEXT

The deck is already **listening**

✅ ON RIGHT NOW

The talk log

Press **Q** for an audience question, **N** for a note.
Each one is stamped with the deck and slide on screen.

- ✓ per-slide dwell time, measured
- ✓ markdown export for the write-up
- ✓ no API key, no mic

🧪 EXPERIMENTAL

The co-presenter

A second window listening through Gemini Live.
Silent unless addressed by name, researching in the background.

- </> **getSlideContext** – what’s on screen
- </> **audienceQuestion** – logs what it hears
- </> live transcript + tool-call trace

⌚ NOT BUILT

The speech annotator

Close the loop: the transcript writes the slide’s take-home prose, from what you actually said.

- transcript × slide context
- → generated **.detail** prose
- → a PR against the deck

🧪 THE PARTS WE ALREADY HAVE

Four voice experiments, looking for a job



ambient_assistant

Always listening, proactive audio, completely silent until you say its name. No wake-word detection — the model decides.

mic · tools



gemini_live · speak

Voice agent with contract-verified tool calling. The arithmetic it does out loud is checked before it's spoken.

verified



voice_docparse

Talk to a document. Voice in, DocParse tools invoked live, answers about DOCX, PPTX, XLSX and PDF.

in-browser



co_presenter.ail

The Gemini Live session driven by AILANG compiled to WASM — which is exactly why the presenter *bridges* to it rather than shipping it.

~39MB wasm

⚠️ HONEST ABOUT THE GAP

What it would **take**

★ WHAT YOU'D GET

- Speaker notes written from the talk you *gave*, not the one you planned
- Every question filed against the slide that caused it, automatically
- Timings measured from a real run, pasted straight back into the playlist
- The deck improves each time you give it. The talk becomes its own data.

⚠️ WHAT'S IN THE WAY

- Tracing stops at the tool call — we want `--emit-trace` inside WASM, which AILANG doesn't expose yet
- A live mic in a webinar is a second thing that can fail, and it's the one you can't hide
- Transcripts of a whole talk are personal data. Whose? Kept where?
- A model that writes your speaker notes will write them *confidently wrong* sometimes

 STEAL THIS

Four steps, deliberately small

01 Write your design system as a page of English.

Your brand colours as hex codes. Two fonts. One rule about what never happens. That's enough to start — it does not need to be complete to be useful.

02 Steal a working deck as the boilerplate.

One that already renders correctly beats a perfect specification. Mine is public; use it, or use any deck whose layout you like.

03 Add exactly one linter rule.

The one mistake you're tired of seeing. Wire it to run after every edit the agent makes. One rule that always fires beats twenty that nobody runs.

04 Ask for a slide. Read it. Say what's wrong.

This is the whole job now. You are the taste, not the typist — and “that heading is too big” is a perfectly good bug report.

⚠ BEFORE YOU GO AND DO THIS

When it's not worth it

📌 THE HONEST VERSION

This is **not less work** than PowerPoint for a three-slide status update on Thursday. Open PowerPoint. I do.

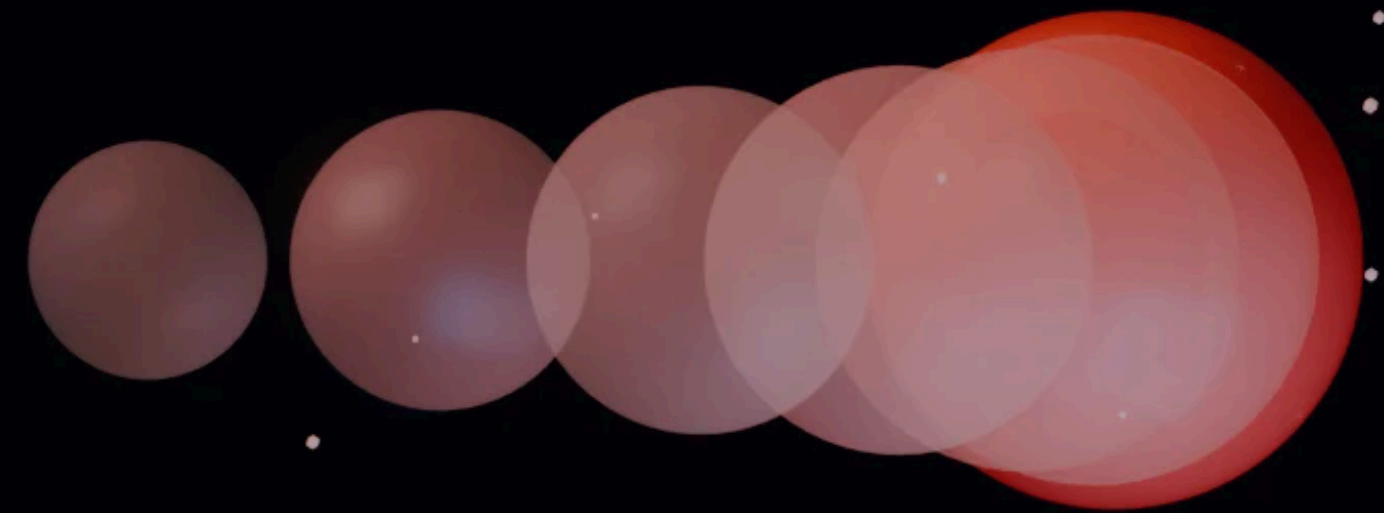
It is **dramatically less work from about talk number three onward** — because that's when the reuse starts paying, and it compounds from there.

And expect the first deck to take **a whole afternoon**. Budget for that, or you'll abandon it an hour in and conclude it doesn't work.

**The tool stopped being the
constraint.**
Ask for **more from a slide.**

 sunholo.com/presentations – every deck, and the template

 the write-up



👋 THANK YOU

Let's keep **talking**



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SUBSTACK
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Questions, ideas, or want to keep the conversation going? *Any of the above works.*